

Modeling 2D Marshak Waves

Understanding the physics of X-ray radiation-driven supersonic nonlinear heat
waves

Yuval Kochavi

Thesis presentation

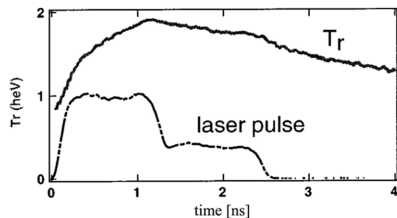
- 1 Marshak Waves Experimental Setup
- 2 Describing the System
- 3 1.5D Model
 - 1D Effects
 - 2D Effects
- 4 Full 2D Model
 - Modeling the Front Curvature
 - Comparing to 2D Supersonic Diffusion Simulation
 - Full 2D Temperature Distribution
 - Comparison with Experimental Data
- 5 Conclusion

About Me

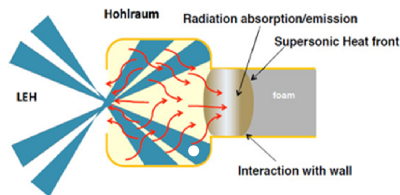
- 19 years old, from Petah Tikva.
- Studied Physics and Engineering at Bar-Ilan University during high school.
- Started research in semester C, after taking courses in Physics, Mathematics, and Computer Science in previous semesters.

Marshak Waves Experiments

Using a Hohlraum as an X-ray generator, a supersonic heat wave propagates in the low-density foam.



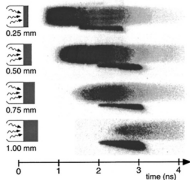
Back et al. POP, (2000)



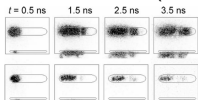
Cohen et al. PRR, (2020)

Laser $\Rightarrow$ Hohlraum $\Rightarrow$ X-ray radiation $\Rightarrow$ supersonic heat wave propagating in the low-density foam.

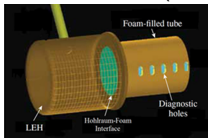
Marshak Waves Experimental Setup



Back et al. POP, (2000)

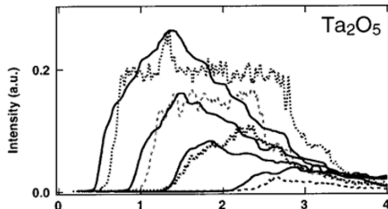


Rosen et al. ASS, (2006)

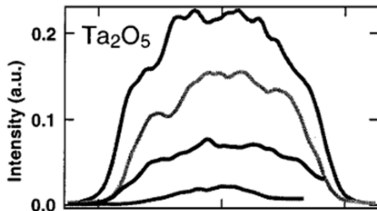


Cohen et al. PRR, (2020)

There are several ways to track the heat wave propagation in experiments:



Back et al. POP, (2000)



Experiments Tested

The experiment	foam type	density (mg/cc)	temperature (eV)
Massen <i>et al.</i>	$C_{11}H_{16}Pb_{0.3852}$	80	100–150
Back <i>et al.</i>	SiO_2 , Ta_2O_5	10–50	85–190
Xu <i>et al.</i>	C_6H_{12} , $C_6H_{12}Cu_{0.394}$	50	160
Ji-Yan <i>et al.</i>	C_8H_8	160	175
Keiter <i>et al.</i>	$C_{15}H_{20}O_6$, $C_{15}H_{20}O_6Au_{0.172}$	45–70	200–210
Moore <i>et al.</i>	SiO_2 , C_8H_7Cl	90–120	310
Courtois <i>et al.</i>	SiO_2	19 – 29	170

The Governing Equations

Scattering-free gray Boltzmann equation for photons (RTE):

$$\frac{1}{c} \frac{\partial I(\hat{\Omega}, \vec{r}, t)}{\partial t} + \hat{\Omega} \cdot \vec{\nabla} I(\hat{\Omega}, \vec{r}, t) = \sigma'_a(T_m) \left(B(T_m(\vec{r}, t)) - I(\hat{\Omega}, \vec{r}, t) \right) \quad (1)$$

Energy exchange between radiation and matter:

$$\frac{C_v(T_m)}{c} \frac{\partial T_m}{\partial t} = \sigma'_a(T_m) \left(\frac{1}{c} \int_{4\pi} I d\hat{\Omega} - a T_m^4 \right) \quad (2)$$

From RTE to Moment Equations

We can define the energy density and flux:

$$E(\vec{r}, t) = \frac{1}{c} \int_{4\pi} I(\hat{\Omega}, \vec{r}, t) d\hat{\Omega}, \quad \vec{F}(\vec{r}, t) = \int_{4\pi} I(\hat{\Omega}, \vec{r}, t) \hat{\Omega} d\hat{\Omega}$$

Now the Boltzmann equation takes the following form:

$$\frac{\partial E(\vec{r}, t)}{\partial t} + \nabla \cdot \vec{F}(\vec{r}, t) = \sigma'_a(T_m) \left(\int_{4\pi} \frac{B(\vec{r}, t)}{c} d\hat{\Omega} - E(\vec{r}, t) \right)$$

$$\frac{1}{c} \frac{\partial \vec{F}(\vec{r}, t)}{\partial t} + \frac{c}{3} \nabla E(\vec{r}, t) + \sigma'_a(T_m) \vec{F}(\vec{r}, t) = 0$$

The Diffusion Approximation

Assuming the change of the flux in time is small ($\frac{1}{c} \frac{\partial \vec{F}}{\partial t} \approx 0$):

$$\vec{F}(\vec{r}, t) = -\frac{c}{3\sigma'_a(T_m)} \nabla E(\vec{r}, t)$$

This turns Boltzmann into the diffusion equation:

$$\frac{\partial E(\vec{r}, t)}{\partial t} - \nabla \cdot \left[\frac{\overbrace{D(T_m)}^c}{3\sigma_t(T_m)} \nabla E(\vec{r}, t) \right] = c\sigma'_a(T_m) \left(\overbrace{\int_{4\pi} \frac{B(\vec{r}, t)}{c} d\hat{\Omega}}^{U_R(\vec{r}, t)} - E(\vec{r}, t) \right)$$

Gray Coupled Equations and LTE

From the diffusion equation, setting $\int_{4\pi} \frac{B}{c} d\hat{\Omega} = aT_m^4$, the coupled gray radiation-matter system reads,

$$\begin{aligned} \frac{\partial E}{\partial t} - \nabla \cdot \left(\frac{c}{3\sigma} \nabla E \right) &= c\sigma(aT_m^4 - E), & E &= aT_R^4 \\ \rho \frac{\partial e}{\partial t} &= c\sigma(E - aT_m^4) \end{aligned}$$

Finally, assuming thermodynamic equilibrium (TE) between the radiation field and the material ($T \equiv T_R = T_m$):

$$\frac{\partial T^4}{\partial t} + \rho \frac{\partial e}{\partial t} = \nabla \cdot \left(\frac{c}{3\sigma} \nabla (aT^4) \right) \quad (3)$$

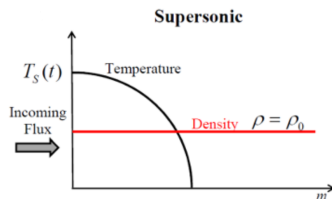
The Problem

- Power-law approximation for specific energy and opacity:

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\sigma} = \frac{1}{\rho\kappa} = gT^\alpha \rho^{-\lambda-1}$$

The Core Challenge

The diffusion coefficient $D = \frac{c}{3\rho\kappa}$ and the material internal energy e are highly nonlinear. The system becomes very difficult to solve analytically.



Shussman et al. POP, (2015)

The Goal

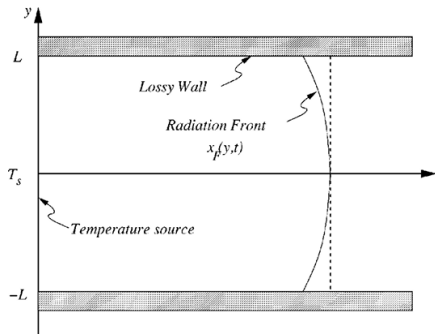
Goal

Build a 2D semi-analytical model to understand supersonic heat wave propagation and temperature distribution.

Specifically, analyzing the following equation:

$$\frac{\partial T^4}{\partial t} + f \rho^{-\mu+1} \frac{\partial T^\beta}{\partial t} = \nabla \cdot (D(T) \nabla (a T^4))$$

using cylindrical coordinates and the relevant boundary conditions.



1.5D Model

HR Model - Governing Equations

Hammer and Rosen solved the diffusion equation with the specific energy and opacity power laws:

$$e = fT^\beta \rho^{-\mu}, \quad \frac{1}{\kappa} = gT^\alpha \rho^{-\lambda}$$

where $f, \mu, g, \lambda, \alpha, \beta$ are material-dependent parameters.

Assuming $\rho f T^\beta \rho^{-\mu+1} \gg a T^4$ (true up to ~ 3 keV), HR solved:

$$f \rho^{-\mu+1} \frac{\partial T^\beta}{\partial t} = \frac{\partial}{\partial x} \cdot \left(\frac{c g T^\alpha \rho^{-\lambda-1}}{3} \frac{\partial}{\partial x} (a T^4) \right)$$

Using perturbation theory, HR derived key physical quantities.

Rosen et al. POP, (2003)

The heat front position $x_F(t)$ is given by:

$$x_F^2(t) = \frac{2 + \varepsilon}{1 - \varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t') dt'$$

where:

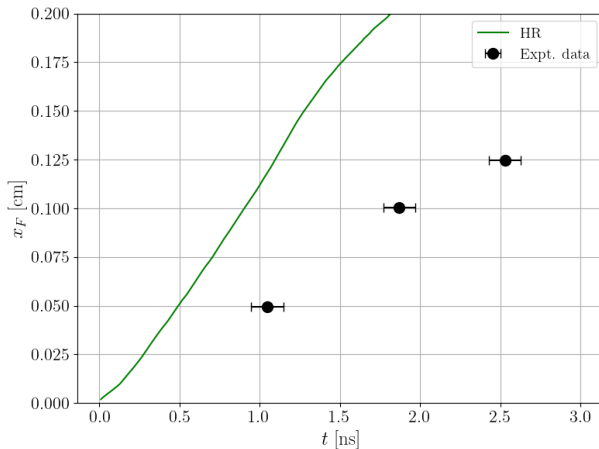
$$C = \frac{16g\sigma_{SB}}{3f(4 + \alpha)} \rho^{2-\mu-\lambda}; \quad \varepsilon = \frac{\beta}{4 + \alpha}$$

$$H(t) = T_s^{4+\alpha}(t)$$

where T_s is the **surface temperature** at $x = 0$.

HR 1D Model Limitations

Naive HR misses the experimental results of Back et al. by a factor of 2!



Conclusion: Assuming $T_s = T_D$ overestimates the wave speed.

Marshak Boundary Condition: $T_D \neq T_s$

The Marshak boundary condition states that

$$\sigma_{SB}T_D^4(t) = \int_0^1 I(\mu)\mu d\mu = \frac{cE(0,t)}{4} + \frac{F(0,t)}{2}$$

where $F(0,t)$ is the net flux entering the foam.

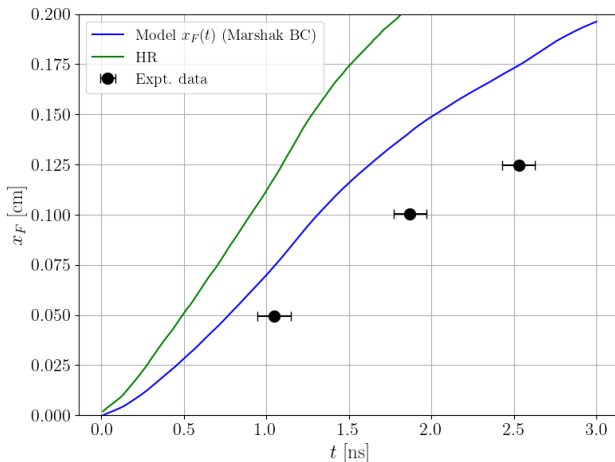
This leads to:

$$\sigma_{SB}T_D^4(t) = \sigma_{SB}T_s^4(t) + \frac{F(0,t)}{2}$$

Therefore, the surface temperature T_s is **always lower** than the drive temperature T_D !

Marshak Boundary Condition: $T_D \neq T_s$

Applying the Marshak boundary condition slows the propagation front significantly:



1.5D Model

2D Effects

Energy Wall Loss - Gold

The heat wave in the foam loses energy to the gold walls, which slows it down:

- Subsonic energy loss of gold per unit area:

$$E_{\text{gold}}(t) = 0.59 T_s^{3.35} [t - t_0(x)]^{0.59} \quad [\text{hJ/mm}^2]$$

where $t_0(x)$ is the front arrival time at position x .

- Energy loss from a tube segment of length dx :

$$dE_W = E_{\text{gold}}(t - t_0(x)) \cdot 2\pi R dx$$

- Total accumulated energy loss in the wall up to time t :

$$E_W(t) = 2\pi R \int_0^{x_F(t)} \int_{t_0(x)}^t \dot{E}_{\text{gold}}(t') dt' dx$$

2D Effect Integration

This energy is subtracted from the foam's energy at each iteration, thereby introducing a two-dimensional effect.

Modeling Other Wall Materials

The same 2D wall loss framework can be applied to other wall materials by using their respective energy loss fits:

- **Copper (Cu) - Mid-Z (subsonic fit):**

$$E_{\text{copper}}(t) = 1.58 T_s^{3.40} [t - t_0(x)]^{0.61} \quad [\text{hJ/mm}^2]$$

- **Beryllium (Be) - Low-Z (supersonic fit):**

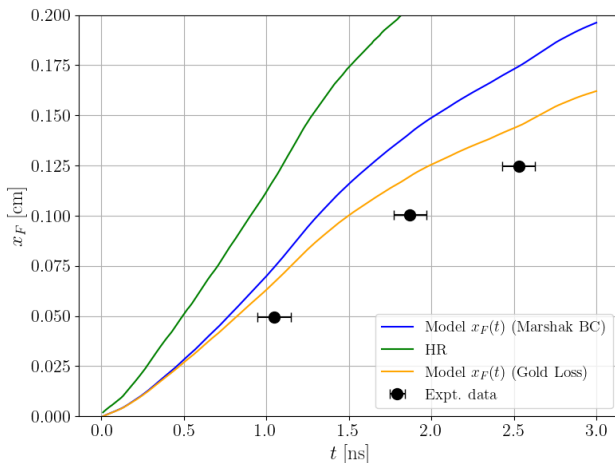
$$E_{\text{be}}(t) = 1.27 T_s^{4.99} [t - t_0(x)]^{0.50} \quad [\text{hJ/mm}^2]$$

- **Vacuum (no external material):**

$$dE_{\text{vacuum}} = \sigma_{\text{SB}} T_s^4 dt \quad (\text{No re-emission from walls})$$

Effect of Wall Loss

Including wall loss reduces the front velocity:



Wall Ablation - Changing Foam Radius

Gold wall material absorbs energy and ablates inward:

- Inward wall velocity:

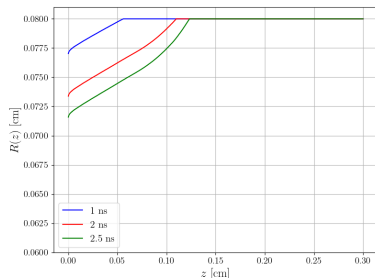
$$u_{\text{abl}} = -510.1 \tilde{u} T_s^{0.716} [t - t_0(x)]^{0.036}$$

- Shrunk cylinder radius:

$$R(t, x) = R_0 - u_{\text{abl}}(x, t)(t - t_0(x))$$

- Reduced inlet energy:

$$E_{\text{in}}(t) = 2\pi\sigma_{\text{SB}} \int_0^t R^2(t', 0) [T_D^4(t') - T_s^4(t')] dt'$$



Wall Ablation - Changing Foam Density

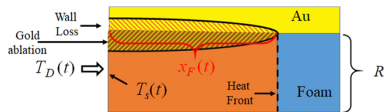
The ablated wall material compresses and densifies the adjacent foam:

- Mass conservation:

$$\rho_{\text{eff}}(t) = \rho_0 \frac{V_0}{V(t)}$$

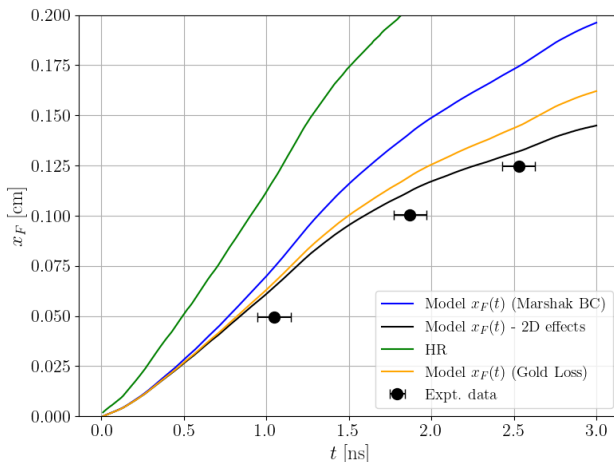
- Effective density:

$$\rho_{\text{eff}}(t) = \frac{\rho_0 V_0}{\int_0^{x_F(t)} \pi R^2(t, x) dx}$$



Effect of Wall Ablation

Inward wall ablation compressively densifies the foam, slowing down propagation significantly:



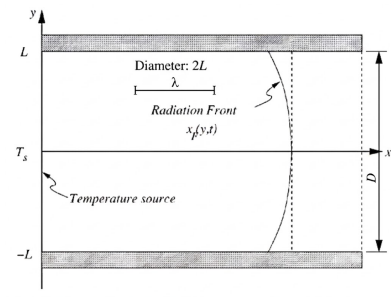
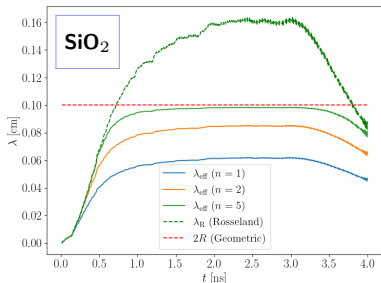
Effective Mean Free Path

When $\lambda_R \geq 2R$, wall collisions control the transport instead of diffusion. This can be modeled as (*Courtois et al. POP, (2024)*):

$$\frac{1}{\lambda_{\text{eff}}} = \frac{1}{\lambda_R} + \frac{1}{2R}, \quad \lambda_R \text{ is the Rosseland mean free path}$$

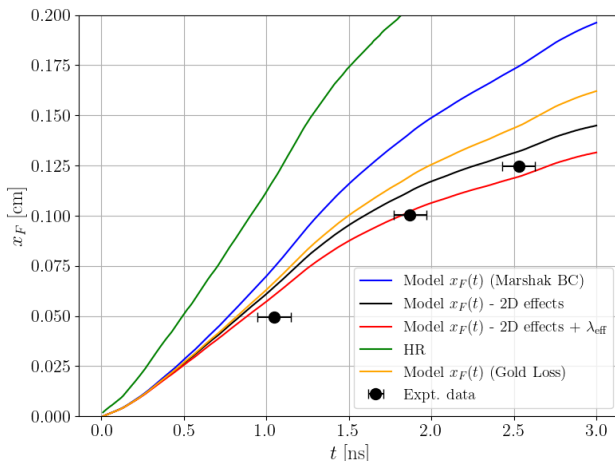
A smoother transition is defined using the generalized power mean:

$$\lambda_{\text{eff}} = \left((2R)^{-n} + \lambda_R^{-n} \right)^{-1/n}$$

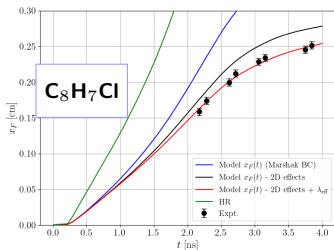


Effect of the Mean Free Path Correction

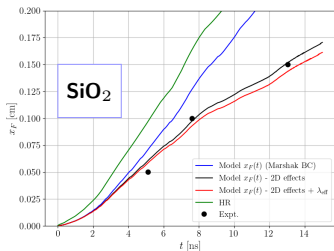
Including the effective mean free path correction matches the model to the experimental data:



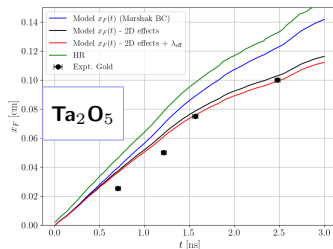
Comparison with Various Experiments



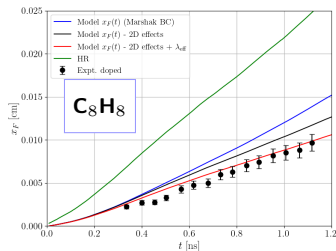
Moore et al. JQSRT, (2015)



Back et al. PRL, (2000)



Back et al. POP, (2000)



Ji-Yan et al. CPB, (2010)

Flux Predicted by the Model

Using the Henyey-like profile:

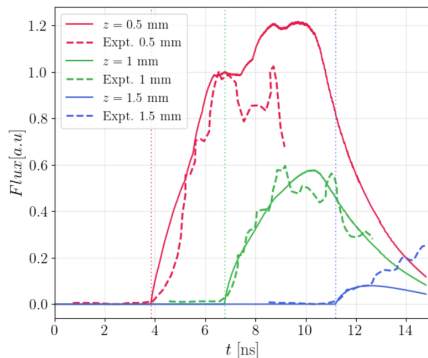
$$T(z, t) = T_S(t) \left[1 - \frac{z}{z_F(t)} \right]^{\frac{1}{4+\alpha-\beta}}$$

behind the heat front ($z < z_F(t)$),
and $T = 0$ ahead of it.

The radiation flux at each detector
position is then computed as:

$$\Phi(t) = \sigma_{\text{SB}} T^4(z, t)$$

where the flux is computed in several
different spatial locations.



2D Model

Full 2D Model

Modeling the Front Curvature

Wall Boundary Condition

Non-ideal walls introduce energy loss: only a fraction α of incoming flux F_{in} is re-emitted ($F_{\text{out}} = \alpha F_{\text{in}}$).

- Net wall flux loss:

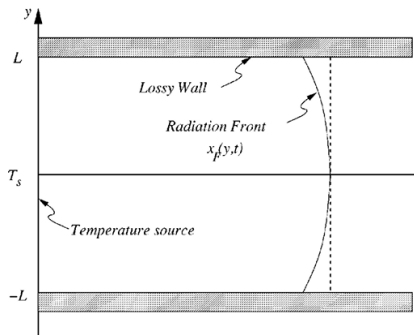
$$F_{\text{wall}} = (1 - \alpha) F_{\text{in}} = (1 - \alpha) \sigma_{\text{SB}} T^4$$

- Equating to diffusion flux (Fick's law):

$$F_{\text{wall}} = -\frac{4}{3} \frac{\sigma_{\text{SB}}}{\rho \kappa} \frac{\partial T^4}{\partial n}$$

- Resulting boundary condition:

$$\left. \frac{\partial T^4}{\partial n} \right|_{\text{wall}} = \frac{3}{4} \rho \kappa (\alpha - 1) T^4 \Big|_{\text{wall}}$$



Understanding the Front Curvature

Hurricane & Hammer (2006) solved the Laplace equation (assuming constant opacity and neglecting e):

$$\underbrace{\frac{\partial e}{\partial t}}_{\rightarrow 0} = \frac{4}{3} \nabla \cdot \left(\frac{1}{\rho \kappa} \nabla (T^4) \right)$$

and got the 2D front in **Cartesian coordinates** (where $\varepsilon = \frac{3}{4} \rho \kappa L (1 - \alpha)$):

$$x_F(y, t) = \frac{L}{\varepsilon} \cosh^{-1} \left(1 + \frac{D \varepsilon t}{L^2} \right) \cos \left(\frac{\sqrt{\varepsilon}}{L} y \right) + \text{HOT}$$

Cylindrical Coordinates

The model was re-developed in **cylindrical coordinates** (*Courtois et al. POP, (2022)*):

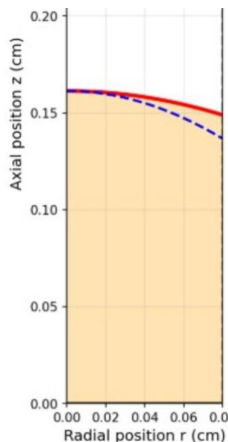
$$x_F(r, t) = \frac{1}{\kappa_0} \cosh^{-1} \left(1 + \frac{D\kappa_0^2 t}{2} \right) J_0(\kappa_0 r) + HOT$$

Planar $\longleftrightarrow$ **Cylindrical Analogy:**

$\cos(ky) \longleftrightarrow J_0(\kappa r)$ For small ε :

$$\kappa_0 \approx \frac{\sqrt{2\varepsilon}}{R}, \quad \varepsilon = \frac{3}{4} \rho \kappa R (1 - \alpha)$$

where α is the wall albedo.



Applying to Our Model: Dynamic Albedo

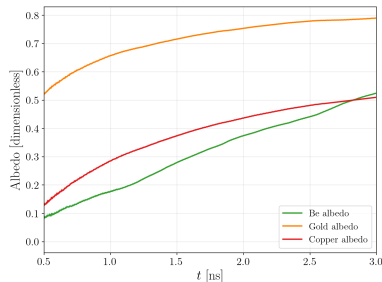
In previous works, the wall albedo was treated as a constant fitting parameter. We update it dynamically over time:

$$\alpha(t) = \frac{F_{\text{out, wall}}}{F_{\text{in, wall}}} = \frac{\sigma_{\text{SB}} T_{s, \text{interface}}^4 - \frac{dE_{\text{wall}}}{dt} / 2}{\sigma_{\text{SB}} T_{s, \text{interface}}^4 + \frac{dE_{\text{wall}}}{dt} / 2}$$

where the front curvature depends on the radial eigenvalue κ_0 :

$$\kappa_0(t) \approx \frac{\sqrt{2\varepsilon(t)}}{R}, \quad \varepsilon(t) = \frac{3}{4} \rho \kappa R (1 - \alpha(t))$$

which dynamically scales with the time-dependent wall loss $(1 - \alpha(t))$.



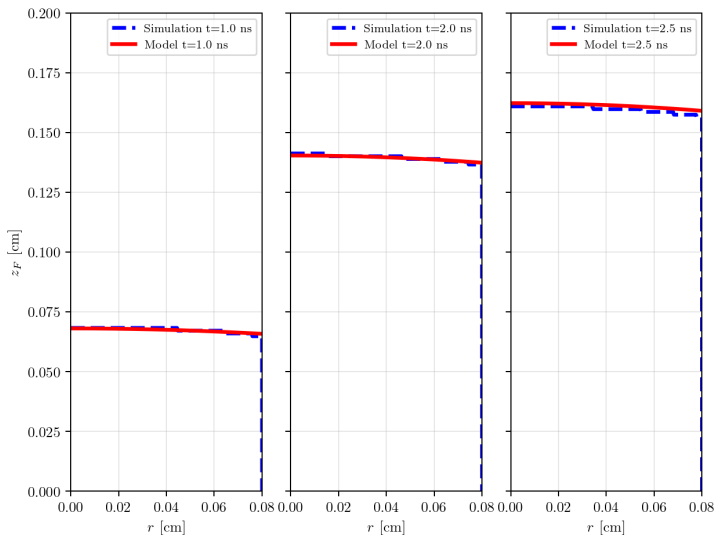
The 2D Wave Front

$$\boldsymbol{x}_F(\boldsymbol{r}, t) = \boldsymbol{x}_{1.5\text{D}}(t) \cdot J_0(\kappa_0(t)\boldsymbol{r})$$

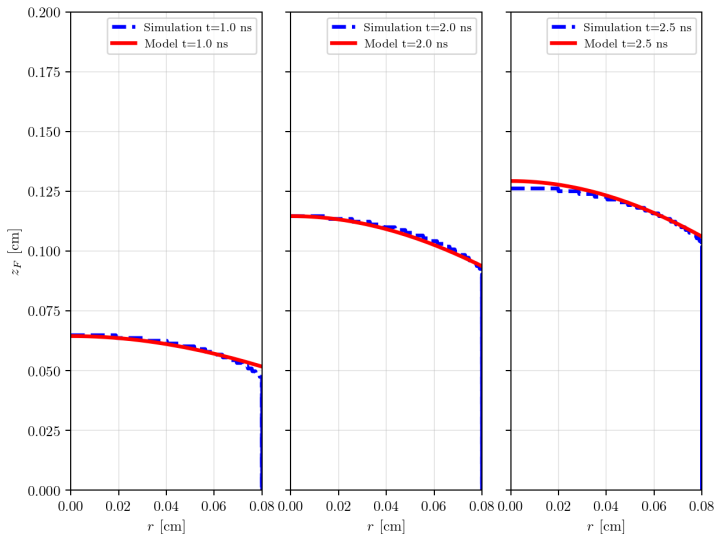
Full 2D Model

**Comparing to 2D Diffusion
Simulation Without Hydrodynamics**

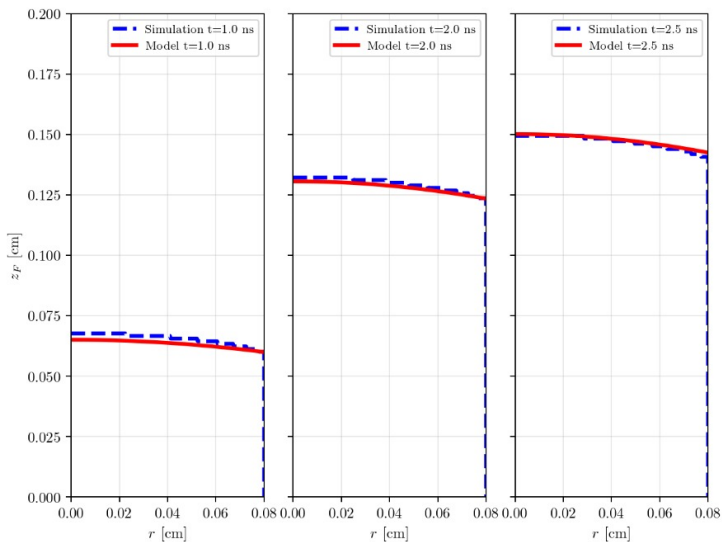
Gold Coating — SiO_2



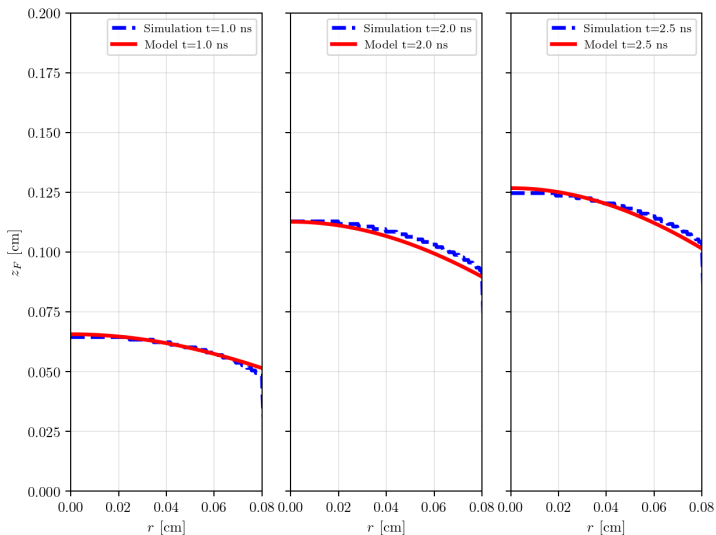
100× Optically Thinner Gold Coating — SiO₂



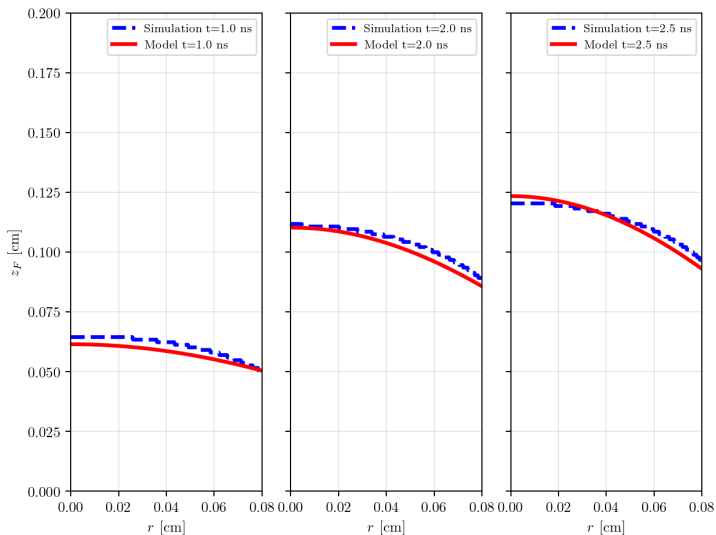
Copper Coating — SiO_2



Beryllium (Be) Coating — SiO_2



Vacuum Boundary — SiO_2



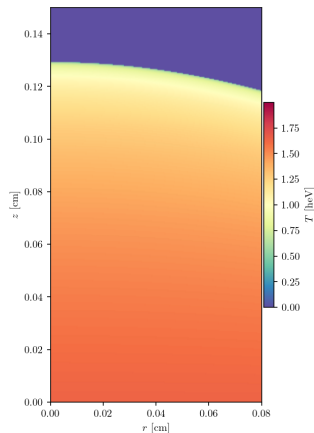
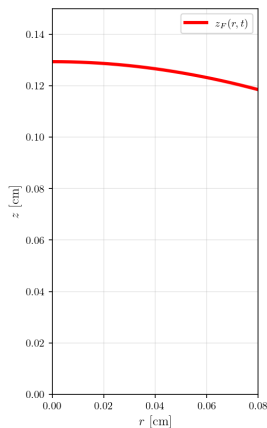
Full 2D Model

A Full 2D Temperature Distribution

Full 2D Temperature Distribution

The 2D Henyey-like temperature profile is given by:

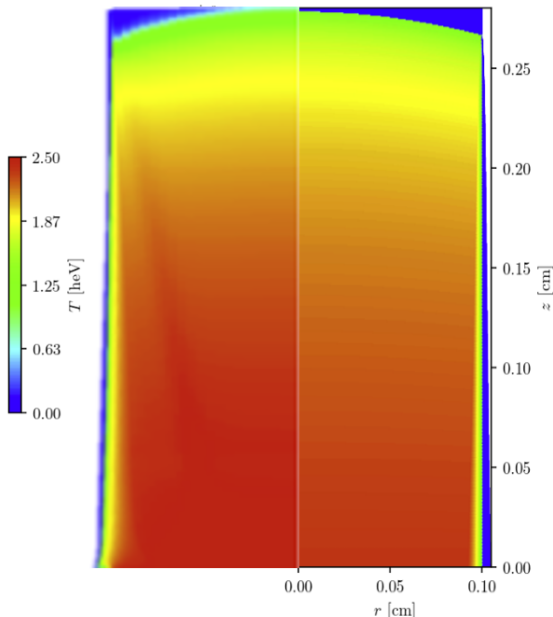
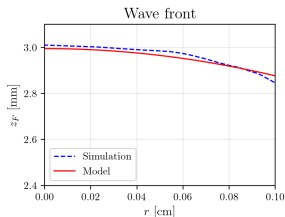
$$T(z, r, t) \approx T_S(t) \left[1 - \frac{z}{z_F(r, t)} \right]^{\frac{1}{4+\alpha-\beta}}$$



2D Temperature Distribution — Moore (C_8H_7Cl)

Side-by-side comparison of the 2D temperature distribution at $t = 3.72$ ns:

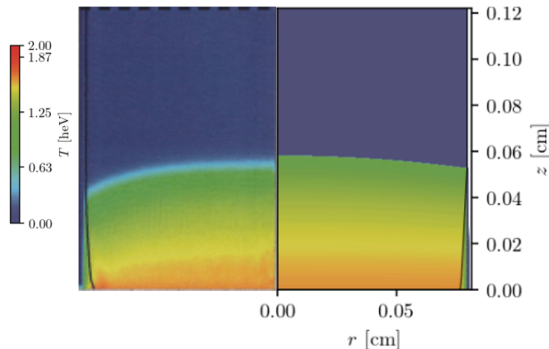
- **Left:** Diffusion simulation with hydrodynamics taken from *Moore et al. JQSRT, (2015)*.
- **Right:** Our full 2D model.



2D Temperature Distribution — Back *et al.* SiO₂ (1.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.0$ ns:

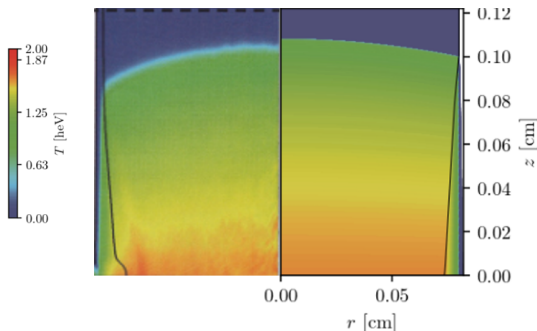
- **Left:** Implicit Monte Carlo (IMC) simulation from *Cohen et al. PRR, (2020)*.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.



2D Temperature Distribution — Back *et al.* SiO₂ (2.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 2.0$ ns:

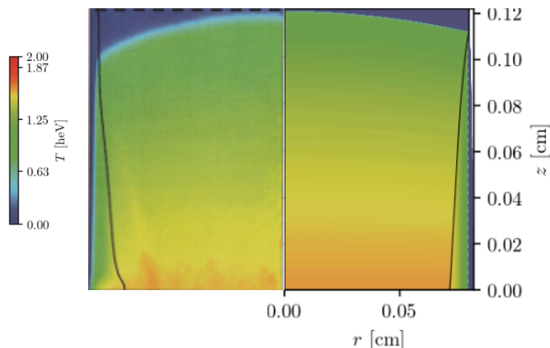
- **Left:** Implicit Monte Carlo (IMC) simulation from *Cohen et al. PRR, (2020)*.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.



2D Temperature Distribution — Back *et al.* SiO₂ (2.5 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 2.5$ ns:

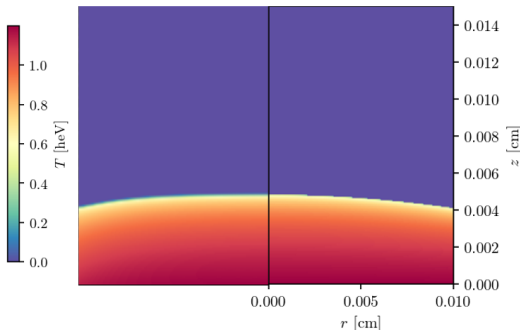
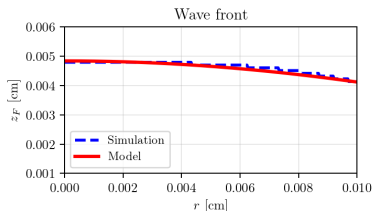
- **Left:** Implicit Monte Carlo (IMC) simulation from *Cohen et al. PRR, (2020)*.
- **Right:** Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.



2D Temperature Distribution — Ji-Yan (0.5 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 0.5$ ns:

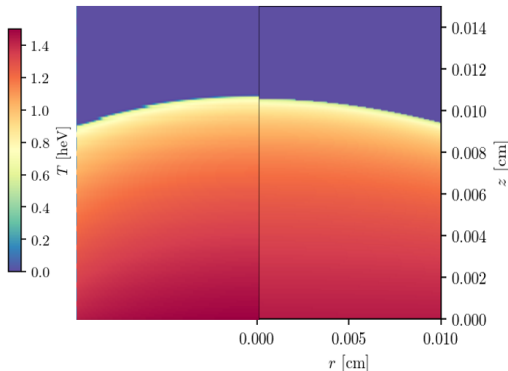
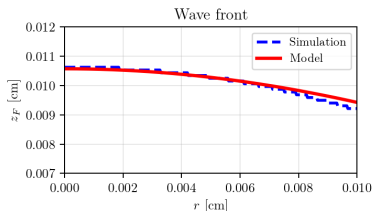
- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



2D Temperature Distribution — Ji-Yan (1.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.0$ ns:

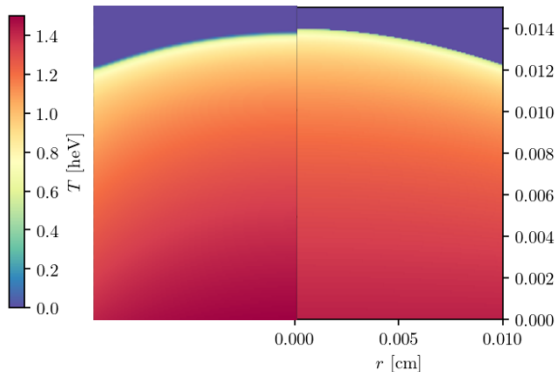
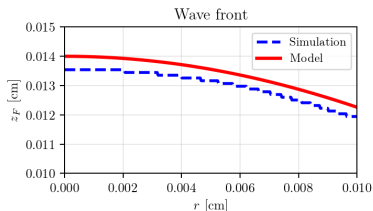
- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



2D Temperature Distribution — Ji-Yan (1.3 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.3$ ns:

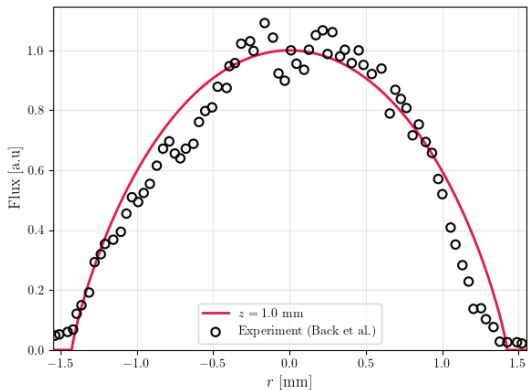
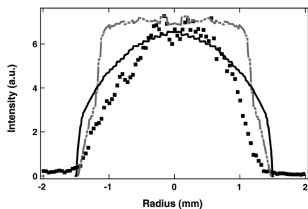
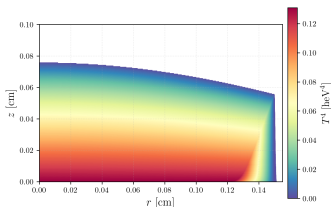
- **Left:** 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right:** Our full 2D model.



Full 2D Model

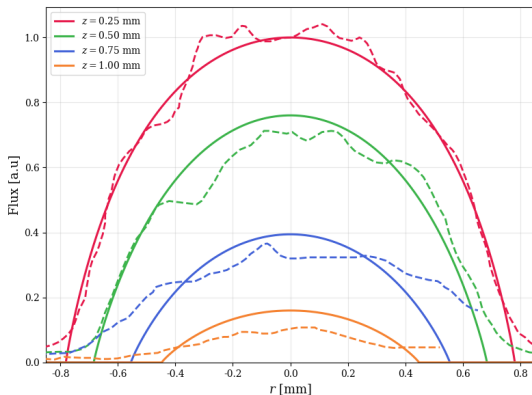
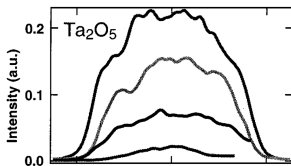
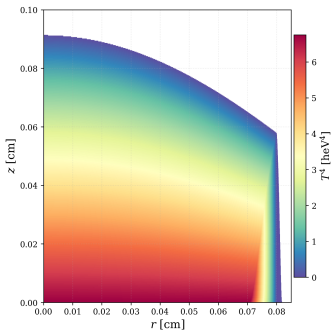
Comparison with Experimental Data

Flux Curvature Comparison — Back *et al.* SiO₂ (PRL)



Back *et al.* PRL, (2000)

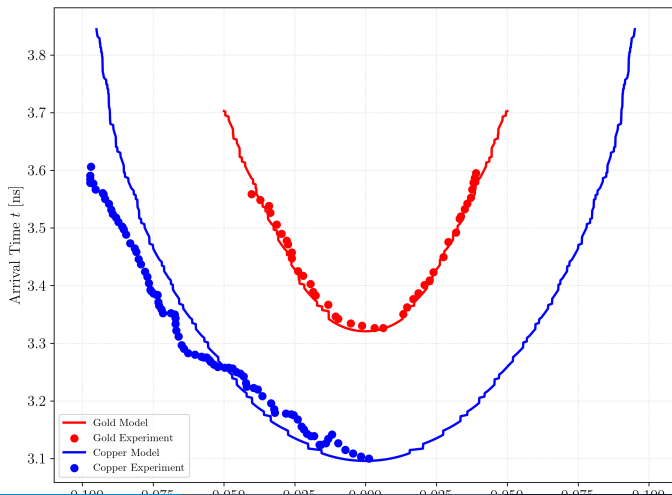
Flux Curvature Comparison — Back *et al.* Ta₂O₅ (POP)



Back *et al.* POP, (2000)

Gold vs. Copper Coating Comparison

Comparison with the French experiment by *Courtois et al. POP, (2024)*, measuring the front arrival time as a function of r for both gold and copper coatings:



- **Successful 2D Modeling:**
 - We have successfully modeled 2D supersonic Marshak waves.
 - Our unified model captures front curvature and temperature distribution under the diffusion approximation.
- **Deep Physical Insight:**
 - We can now understand the underlying physics of radiation transport and energy exchange in an intuitive and clear manner.

Thank You!

Questions?